Curriculum Vitae

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Yiwen Ding

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Research Interests

My research lies in category theory and theoretical computer science, with particular interests in logic in computer science, categorical algebra, universal algebra, coalgebra, and the logical foundations of artificial intelligence.

Education

- **PhD Candidate in Logic and Mathematics** Sep. 2022 – present
  Vrije Universiteit Amsterdam, The Netherlands
  Supervisors: Alessandra Palmigiano and Alexander Kurz
  Doctoral thesis submitted; expected graduation by the end of 2026.
- **MA in Philosophy** Sep. 2019 – Jun. 2022
  Shandong University, China
  Advisor: Fei Liang
- **BA in Law** Sep. 2015 – Jun. 2019
  Qufu Normal University, China

Doctoral Thesis

Quantale-Enriched Algebras

Develops a theory of algebras enriched over a commutative quantale, connecting classical universal algebra with ordered and quantitative settings. Using enriched category theory, the thesis establishes a congruence–quotient correspondence and proves Birkhoff-style and Pavelka-style completeness for a graded inequational logic. It proves enriched SP and HSP theorems characterizing classes of algebras definable by implications and inequations, and transports these results to separated algebras via separated reflection. These results provide a common approach to algebraic reasoning with equality, order, and generalized distance.

Publications

Journal Articles

1. Ruoding Wang, **Yiwen Ding**, Krishna Manoorkar, Apostolos Tzimoulis, Xiaolong Wang. Causal Kripke models. *Synthese 206(3), Article 133, pp. 1–36 ; 2025.* DOI: 10.1007/s11229-025-05223-z.

Refereed Conference Articles

1. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni, Apostolos Tzimoulis. Hybrid Logics with Moving Names. *Dynamic Logic. New Trends and Applications (DaLí 2025) ; LNCS 16472, pp. 23–39 ; 2026.* DOI: 10.1007/978-3-032-22626-6_2.
2. **Yiwen Ding**, Krishna Manoorkar, Apostolos Tzimoulis, Ruoding Wang. Probabilistic Causal Kripke Models. *Logic, Rationality, and Interaction (LORI 2025) ; LNCS 16010, pp. 51–64 ; 2026.* DOI: 10.1007/978-981-95-2481-5_4.

3. **Yiwen Ding**, Krishna Manoorkar, Alessandra Palmigiano, Ruoding Wang. Monotone Modal Logic Beyond Distributivity. *Logic and Its Applications (ICLA 2025) ; LNCS 15402, pp. 111–125 ; 2025.* DOI: 10.1007/978-3-031-89610-1_8.
4. **Yiwen Ding**, Krishna Manoorkar. Fuzzy Lattice-based Description Logic. *Electronic Proceedings in Theoretical Computer Science 421, pp. 44–63 ; 2025.* DOI: 10.4204/EPTCS.421.3.
5. **Yiwen Ding**, Krishna Manoorkar, Mattia Panettiere, Ruoding Wang. Toward the van Benthem Characterization Theorem for Non-Distributive Modal Logic. *Advances in Modal Logic (AiML 2024) ; 2024.* arXiv:2404.05574.
6. **Yiwen Ding**, Krishna Manoorkar, Apostolos Tzimoulis, Ruoding Wang, Xiaolong Wang. Causal Kripke Models. *Electronic Proceedings in Theoretical Computer Science 379, pp. 185–200 ; 2023.* DOI: 10.4204/EPTCS.379.16.

Conference Abstracts

1. **Yiwen Ding**, Alexander Kurz. Quantale-Enriched Universal Algebra: Completeness. *Booklet of Abstracts, TACL 2026, p. 161 ; 2026.* Abstract.
2. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni. Multi-type universal algebra: Transfer of properties. *TACL 2024 ; Conference abstract.* Abstract.
3. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni. Multi-type universal algebra: categorical equivalence. *TACL 2024 ; Conference abstract.* Abstract.

Preprints and Manuscripts

1. **Yiwen Ding**. Preservation and definability for the fluted fragment. *2026; under review.* arXiv:2607.12970.
2. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni, Ruoding Wang. Defeasible Reasoning on Concepts. *2024; under review.* arXiv:2409.04887.
3. **Yiwen Ding**, Krishna Manoorkar, Mattia Panettiere, Apostolos Tzimoulis, Ruoding Wang. Game semantics for lattice-based modal μ-calculus. *2023, revised 2024; under review.* arXiv:2310.13944.

Formalization

Modal Characterization in Lean 4 2026

Lean 4 formalization of classical modal model theory, built on mathlib and CSLib. Includes van Benthem's characterization theorem for basic unimodal logic, modal definability of pointed model classes, bisimulation-somewhere characterizations, and Hennessy–Milner results, together with supporting saturation and ultraproduct constructions. The repository provides theorem-type checks, axiom audits, and continuous integration.

Teaching Experience

- **Teaching Assistant — Logic, Information, Argumentation** Fall 2024
  Amsterdam University College, The Netherlands. Undergraduate course.
- **Teaching Assistant — Modal Logic**
  Shandong University, China. Undergraduate course; during MA studies.
  Course instructor: Chao Xu

Talks and Conference Presentations

1. Topology, Algebra, and Categories in Logic (TACL 2026). 2026. Abstract.
2. *An Invitation to Quantale-Enriched Universal Algebra*. GALAI Seminar, Chapman University, USA. April 29, 2026. Abstract.
3. Dynamic Logic. New Trends and Applications (DaLí 2025). 2025. Paper abstract.

4. 7th Asian Workshop on Philosophical Logic (AWPL 2025). Kolkata, India, 2025. Programme.
5. ESSLLI 2024 Student Session. Leuven, Belgium, 2024. Programme.
6. Advances in Modal Logic (AiML 2024). Prague, Czech Republic, 2024. Abstract.
7. Topology, Algebra, and Categories in Logic (TACL 2024). Barcelona, Spain, 2024. Abstract.
8. MOSAIC Workshop 2023: Modalities in Substructural Logic II – Theory, Methods and Applications. TU Wien, Vienna, Austria. September 27–29, 2023. Programme.

Academic Visits

- Chapman University, USA
  Center of Excellence in Computation, Algebra, and Topology (CECAT)
  Mar. 2026; Oct. 2025 — Host: Alexander Kurz
  May 2024; Apr. 2023 — Host: Peter Jipsen
- University of the Witwatersrand, South Africa Jan. 2023
  School of Mathematics; host: Willem Conradie

Scholarships and Support

China Scholarship Council: full funding for PhD studies in the Netherlands (2022–present).